EXPERIENCEBRIDGE

One robot’s recovery. Another robot’s next decision.

An evidence-backed exchange for robot recovery lessons.

THE WEDGE

Share a compatible recovery before another machine repeats the failure.

THE PROOF

A working, deterministic three-arm simulator with 15 passing tests.

VERIFIED: npm run demo; npm test, 13 October 2026.

01 / PROBLEM

Useful recovery lessons need a path across owners.

Start with connector insertion across independent factories.

CURRENT HYPOTHESIS

Operators may repeat commissioning and recovery work when lessons stay inside deployments.

WHAT MUST TRANSFER

Hardware context, evidence and permission to use the lesson. A procedure alone is insufficient.

INFERRED: commercial hypothesis. Source: vision.md, Why now / problem.

Consult experience before execution.

A narrow path from a recorded failure to a compatible advisory.

CONTRIBUTE

Arm A records a failure and a simulator-tested recovery.

CONSULT

Arm B queries by task, hardware and fixture. The service returns an advisory only when the gates pass.

REJECT

Arm C has a different gripper. The service refuses the transfer.

VERIFIED: scripts/demo.ts; src/memory/match.ts; npm run demo.

Same fixture.
Different decision.

Constructed scenario: seed 7, fixture-7, connector insertion.

WITHOUT MEMORY

Arm B: 12 N → slip. Baseline failures: 1/1.

WITH MEMORY

Arm B: 8 N → success. Recovery successes: 1/1.

INCOMPATIBLE ARM

Arm C: vacuum-v2 → gripper mismatch. Advisory: null.

VERIFIED: npm run demo. One fixed case; no statistical or real-world performance claim.

A useful lesson needs boundaries.

The service refuses unsafe or unsupported recommendations.

COMPATIBILITY

Task, robot model, gripper, connector, material, fixture and force envelope must match.

EVIDENCE & LIFECYCLE

Replay the fixed recovery. Reject expired, revoked or insufficient-evidence records. Missing context requires review.

INTEGRITY

Canonical SHA-256 detects edits against the saved digest. Contributor claims remain unauthenticated.

VERIFIED: src/memory/match.ts; src/memory/integrity.ts; npm test.

The traded service: a permitted recovery lesson.

Proposed deal: Factory B licenses an applicable insight from Factory A.

VALUE EXCHANGE

Contributor supplies a lesson with evidence and permitted uses. Consumer obtains a compatible advisory.

PEAQ ROLE

Planned machine identity links attribution to machines. Planned settlement records licensed access.

DATA BOUNDARY

Keep experience content offchain. Record economic events without publishing proprietary telemetry.

INFERRED: proposed economic design. VERIFIED: current receipt is simulated; transactionHash: null.

The core path runs today.

TypeScript + Fastify + Zod + local JSON + Node crypto.

WORKING

Publish → persist → query → verify → advise → replay. HTTP API and one-command CLI demo.

CHECKED

15 tests passed, 0 failed. Includes mismatches, tampering, expiry, unsafe force and simulated payment errors.

INTEGRATION STATUS

peaq receipt: simulated. Machine ID: synthetic. dstack and ROS: not implemented.

VERIFIED: npm test: tests 15, pass 15, fail 0. Source: package.json; README.md.

Test the cross-owner exchange.

The proposed value is compatible transfer with explicit rights.

EXISTING STACK

Controllers handle execution. Simulators test procedures. Fleet tools already capture operational knowledge.

EXPERIENCEBRIDGE THESIS

A context-gated exchange across independent owners may add value to those systems.

HOW TO EARN THE CLAIM

Test held-out scenarios and interview operators. Compare against existing fleet recovery workflows.

INFERRED: positioning and differentiation hypothesis. Source: vision.md, competitive risks.

Prove usefulness beyond the toy scenario.

Advance only when technical and commercial evidence support it.

NEXT TECHNICAL PROOF

Activate a real peaq machine identity and verify a permitted activity or settlement event.

NEXT TRANSFER PROOF

Use held-out tasks and multiple configurations. Measure correct transfers and rejected mismatches.

NEXT BUYER PROOF

Seek two independent operators and test whether avoided recovery work justifies integration and licensing costs.

INFERRED: proposed next work. Source: vision.md, Decision gates. No pilots or revenue claimed.

09 / BOUNTY FIT

A working prototype for machine knowledge commerce.

Built for the peaq Machine Economy track.

TECHNICAL IMPLEMENTATION

Reproducible demo, typed API, deterministic gates and negative tests.

ECONOMIC FOCUS

A machine-linked insight becomes a proposed licensed service between owners.

HONEST SUBMISSION

Show the working demo first. Disclose simulated economics and the missing live peaq integration.

VERIFIED: bounty criteria fetched 13 October 2026. See speaker notes for listing URL.

Help machines reuse a recovery worth sharing.

ExperienceBridge proves one narrow transfer path today.

WATCH THE PROOF

npm run demo

WHAT WE WANT TO VALIDATE

Real machine attribution, permitted exchange and transfer beyond one constructed case.

VERIFIED: working local simulator. INFERRED: future exchange value.